

Printer compatibility and setup guide

This document describes which printers work with the React label printer web app, how printing is routed through your operating system, and practical steps to get a good print on real label stock.

How compatibility works

The app renders a label in the browser and opens the standard print dialog. Your OS and printer driver send the job to the printer. No vendor SDK is required for this workflow.

- Compatible: Thermal label printers with reliable drivers and a paper/media size that matches your label roll.
- Generally avoid relying on generic drivers if the manufacturer provides an official driver or preset for your rolls.

Default label dimensions in the app

The project defaults to 62 mm wide by 29 mm tall (narrow office / address style die-cut labels). Brother QL DK rolls in this size are a common match. If your physical labels differ, update the dimensions in `src/config/labelDimensions.ts` and `src/styles/label-print.css` together.

Recommended hardware (office / SMB)

For the default 62 x 29 mm layout, a Brother QL series printer plus matching DK stock is usually the least ambiguous choice: broad roll catalog, crisp output, and solid macOS and Windows drivers.

- USB-only models suit a single workstation; Wi-Fi or Ethernet models suit shared desks.
- DYMO LabelWriter models can work the same way via the OS print path; verify the roll size matches your CSS before buying.

Brother QL family

- Pros: Large label assortment, sharp thermal output, common in SMB.
- Networking: Compare USB-only vs Wi-Fi/Ethernet if multiple computers will print.
- Brother b-PAC / SDK is optional; not required for browser print dialog flows.

DYMO LabelWriter class

- Pros: Widely used for address-style labels.
- Confirm each SKU supports the label sizes you purchase; check drivers for every OS you use.
- DYMO Connect and JS APIs exist if you later want tighter desktop integration.

Choosing one printer SKU

1. Pick label stock first (exact width x height drives printer compatibility).
2. Choose USB vs Wi-Fi/Ethernet based on how the printer will be shared.
3. Validate drivers on each target OS (especially macOS).
4. If you print barcodes, test scan quality on hardware after any template changes.

Out of scope for this app version

Silent printing without a dialog, raw USB-from-browser printing, and industrial ZPL-over-network printing require different architectures (local agents, WebUSB, or print servers). This project targets dialog-based OS printing only.

Steps: print a label from the web app

1. Install dependencies and start the dev server: `npm install && npm run dev`.
2. Open the local URL shown in the terminal (often `http://localhost:5173`).
3. Fill in title and body fields; confirm the on-screen preview.

4. Click Print label.
5. In the system dialog, choose your label printer.
6. Set paper or media size to match the label (62 x 29 mm or the driver preset for that roll).
7. Disable headers and footers in the browser print options when available.
8. Print a test sheet and confirm alignment on die-cut stock before production use.

Print dialog troubleshooting

- Wrong printer selected: re-open the dialog and pick the thermal label device explicitly.
- Clipping or wrong scale: adjust media size; try Chrome or Edge first (best support for custom page sizes).
- Non-printable margins: tweak driver settings or slightly reduce padding in `src/styles/label-print.css`.

Browser notes

Chromium browsers (Chrome, Edge) typically honor custom page sizes most consistently. If Safari or Firefox scales oddly, retry in Chrome/Edge; exporting to PDF first is a common fallback.

Project files reference

- `src/App.tsx` — form state and Print label action.
- `src/components/LabelPreview.tsx` — label markup.
- `src/styles/label-print.css` — screen styles, `@media` print, `@page` size.
- `src/config/labelDimensions.ts` — label width and height constants.

Generated for the react-label-printer project. Output path: `docs/printer-compatibility-and-setup.pdf`